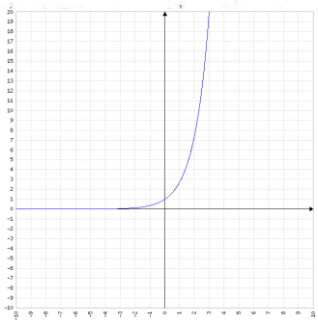
**EXERCÍCIOS**

1) Determine os pontos críticos das funções dadas a seguir. Identifique os intervalos onde f é crescente e decrescente. Determine os extremos locais da função.

a) $f(x) = (x^2 - 3)e^x$

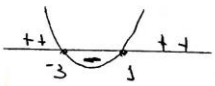
$f'(x) = 2x e^x + (x^2 - 3) e^x = e^x (x^2 + 2x - 3)$

i) $e^x > 0$ p/ $\forall x \in \mathbb{R}$

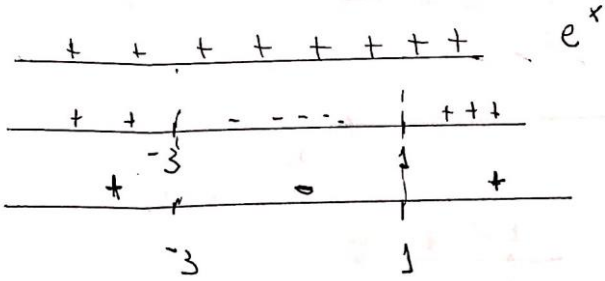


ii) $x^2 + 2x - 3 = 0$

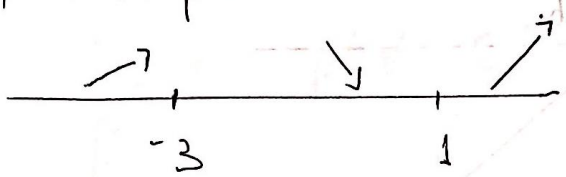
$x = 1$ e $x = -3$




Estudo do sinal de f'




Logo, o comportamento da f é:



$\therefore$

Pontos críticos: $x = -3$ e $x = 1$

f é crescente em $]-\infty\ -3[\ \cup\ ]1\ +\infty[$

f é decrescente em $]-3\ \ 1[$

$x = -3$ é ponto de máximo local

$x = 1$ é ponto de mínimo local

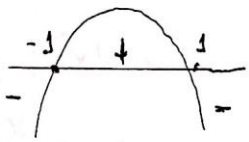
b) $f(x) = \dfrac{x}{x^2+1}$

$$f'(x) = \frac{1(x^2+1) - x(2x)}{(x^2+1)^2} = \frac{-x^2+1}{(x^2+1)^2}$$

i) $-x^2 + 1 = 0 \quad (-1)$

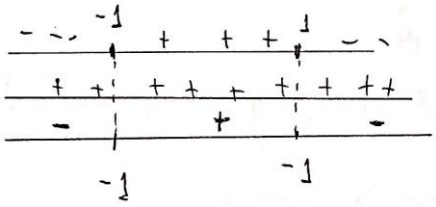
$x^2 - 1 = 0$

$x = \pm 1$

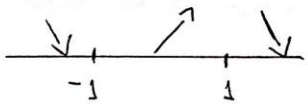



ii) $(x^2+1)^2 > 0$ p/ $\forall x \in \mathbb{R}$

Estudo do sinal de $f'$:




Comportamento da f:



$\therefore$

Pontos críticos: $x = -1$ e $x = 1$

f é decrescente em $]-\infty\ -1[\ \cup\ ]1\ +\infty[$

f é crescente em $]-1\ \ 1[$

$x = -1$ é ponto de mínimo local

$x = 1$ é ponto de máximo local.

2) Esboçar o gráfico da função $f(x) = x^4 + 4x^3$

i) $f'(x) = 4x^3 + 12x^2 = 4x^2(x+3)$

As raízes de $f'(x)$

$4x^2(x+3) = 0 \Leftrightarrow 4x^2 = 0$ ou $x + 3 = 0$

- $4x^2 = 0 \Leftrightarrow (x = 0)$
- $x + 3 = 0 \Leftrightarrow x = -3$

As raízes são $x_1 = 0$ e $x_2 = -3$

ii) $f''(x) = 12x^2 + 24x$

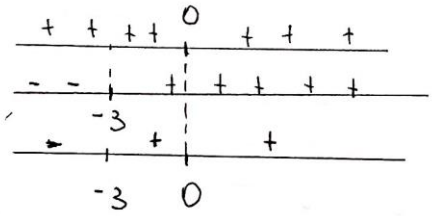
- $f'(0) = 0$, $f''(0) = 0$ (NSPC)

(nada se pode concluir)

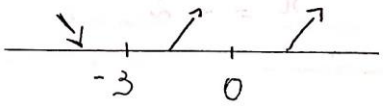
- $f'(-3) = 0$, $f''(-3) = 36 > 0$

$\therefore$ $x = -3$ é ponto de mínimo local

Estudar o sinal de $f'$




Comportamento da $f$



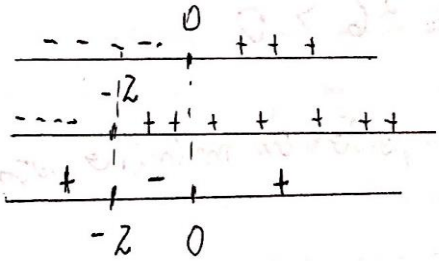

Portanto, em $x = 0$ não temos mínimo nem máximo !!

Estudar o sinal $f''(x)$ para analisar a concavidade

$$f''(x) = 12x^2 + 24x = 12x(x+2)$$

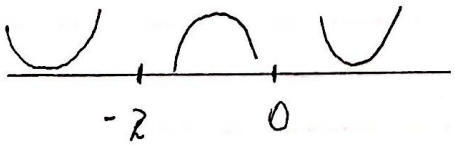
$$12x(x+2) = 0 \Leftrightarrow 12x = 0 \text{ ou } x + 2 = 0$$

- $12x = 0 \Leftrightarrow x = 0$
- $x + 2 = 0 \Leftrightarrow x = -2$




sinal $f''$

Concavidade f




∴ Pontos de inflexão são

$x = -2$ e $x = 0$

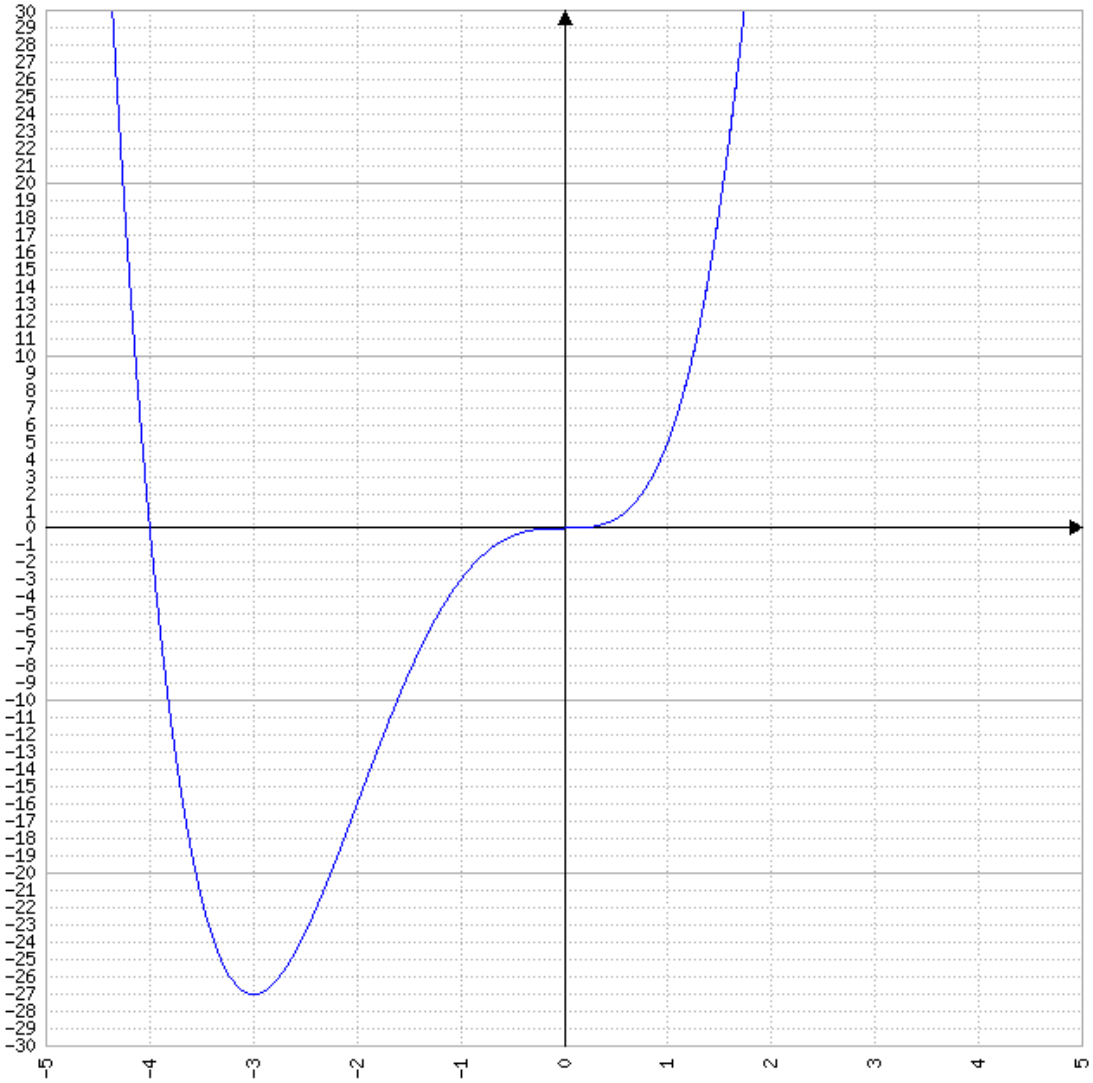
Vamos calcular os limites da função quando $x \to \pm\infty$

$$\lim_{x\to+\infty} x^4 + 4x^3 = \lim_{x\to+\infty} x^3(x + \frac{4}{x}) = +\infty$$

$$\lim_{x\to-\infty} x^4 + 4x^3 = \lim_{x\to-\infty} x^3(x + \frac{4}{x}) = +\infty$$

**Dica para esboçar o gráfico**: calcular $f(x)$ para os pontos de Máximo e mínimo locais, pontos de inflexão e eventuais outros que sirvam para guiar o desenho.

| x | $f(x)$ |
|---|---|
| -3 | -27 |
| -2 | -16 |
| 0 | 0 |
| -4 | 0 |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